Introduction

This document offers the general framework for the Applied Research efforts being proposed for fulfilling a business objective.

Applied research process

The General framework provides means to the lab setup, tools identification and prepare the report for given opportunity for an innovative product. This is NOT a technical document that will be part of the actual product implementation. This research is to demonstrate the key productivity, feature or improvement in a lab setup.

	Basic/foundational/academic Research	Applied Research
Purpose	To advance human understanding and knowledge of the universe. To uncover universal laws, e.g. pertaining to space, matter, energy, society, living and nonliving systems, etc.	To help clients (e.g. policy makers or organizational leaders) to make a decision about a particular situation, problem, or opportunity.
Initiative	Researcher initiates based on their own expertise, research skills, and interest.	Client initiates based on their need for information in response to a particular situation, problem, or opportunity.
Funding	Funding is usually available from government, universities, and private foundations in the form of research grants.	The client pays for a consultant or employs an in-house member of staff. Funding is usually much less than needed for a thorough research.
Who does research	A solo researcher, usually from one discipline Team projects are less common	In-house member of staff or consultant(s). (Ideally it should be a research team consisting of experts from relevant and complementary disciplines.)
Timeframe	Usually longer timeframes compared to applied research	Tied to the client's timeframes; Significantly shorter deadlines than researchers would ideally need.
Research setting	Research can be autonomous from its environment	Research is embedded in, and inseparable from, its real world context
Research methods	Usually uses fewer varieties of data sources and data collection methods. Data collection and analysis methods usually build on the researcher's unique strengths.	Tends to use a combination of multiple methods of data collection from different sources and a mix of qualitative and quantitative data analysis methods.
Evaluation and outlet	Presented in scientific conferences and published in journals subject to peer review process.	Presentation and report submitted only to the client who is the sole evaluator of the work. The client decides whether and how to use the information and whether to make it public.

Fig 1: Characteristics of Basic vs. Applied Research

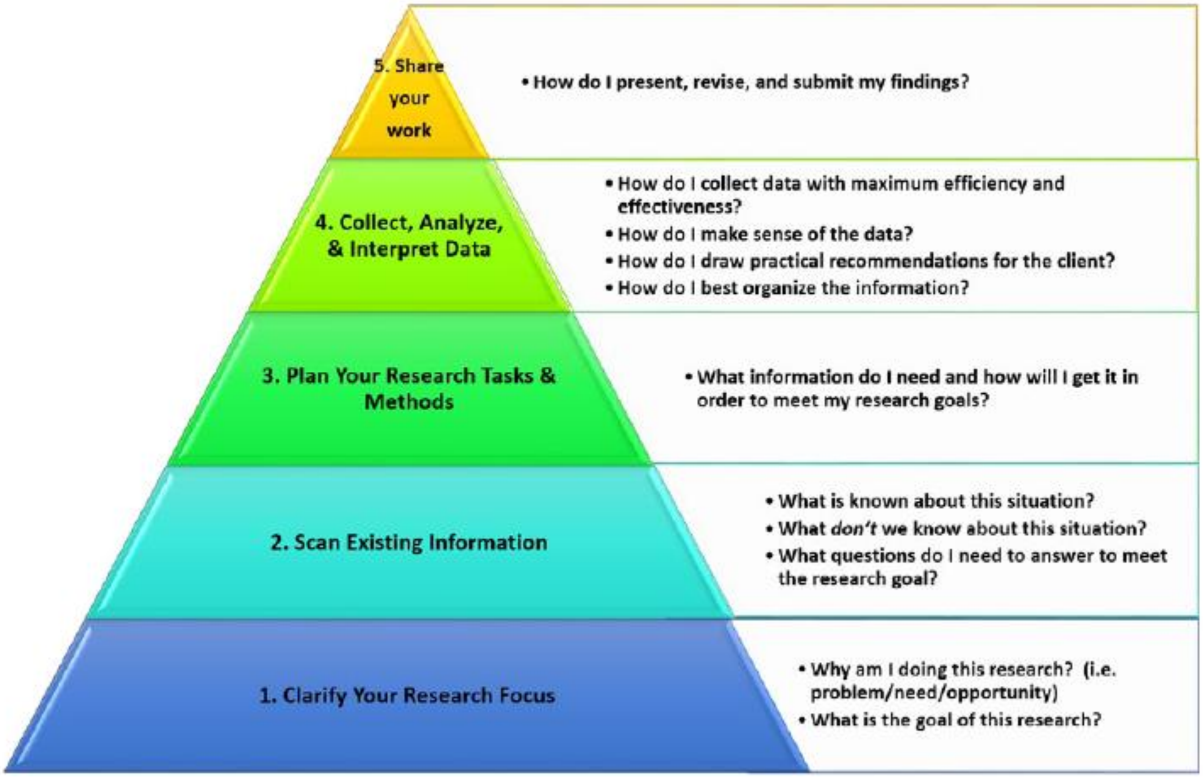

Fig 2: Five-Step Framework for Applied Research Process [1]

Project Goal ▶	Main Research Question
To develop recommendations for the Dean to better align the school's curriculum with core competencies essential for our alums' successful employment in top organizations they wish to work for.	How might the school improve its curriculum to better prepare successful professionals for jobs at top organizations they wish to work for?
Objectives ▶	**Subordinate Questions**
To identify core competencies essential for successful careers in organizations that most students aspire to work for.	Which core competencies account for successful careers in those top organizations for entry-level professionals? Why?
To identify gaps between employers' needs and what the school offers as well as the reasons why this gap exists.	Which of these core competencies sought by top employers are covered by the school curriculum? How effective is the school in helping students to learn the competencies which are being offered? What does it do well? What does it not do so well? Which of these core competencies are missing in the school curriculum? Why?
To generate recommendations for school administration on (1) modified curriculum content and (2) pedagogy to further improve the school curriculum in order to prepare professionals for the needs of top organizations that the school graduates wish to work for.	How might the school change its curriculum content to better align it with the needs of top employers (content question)? How might the school help students to master these core competencies (process/pedagogy question)?

Fig 3: Example of Converting Research Goals into Research Questions

Plan Your Research Tasks and Methods

1. Select the research design framework (experimental or descriptive).
2. Specify the information needed to answer each research question.
3. Identify data sources for each piece of information needed.
4. Specify how you will sample your data sources.
5. (Specify variables and measurements if doing a quantitative study).
6. Select data collection methods.
7. Design data collection instruments/forms/questionnaires and procedures corresponding to each data collection method if you plan on using interviews, surveys, focus groups, or observations. Pilot your data collection instruments and revise them as needed.
8. Identify data analysis methods and tools for qualitative and/or quantitative data as relevant. Specify data collection, coding, and analysis procedures.
9. Plan how you will address validity and reliability issues.
10. Develop a schedule (e.g. using a Gantt chart) for carrying out the tasks listed in this step.
11. (If doing formal proposal add a budget and information about the research team's qualifications.).

Objectives	Questions to answer (or activities needed) to meet your objectives	What information is needed to answer this question	Best source of this info	How will you sample from this source (sampling method)	How will you collect the data (i.e. data collection method)	How will you analyze the data (data analysis method(s))
Objective #1	Question #1					
	Question #2					
Objective #2	Question #3					

Fig 4: Methodology Matrix

Select a Research Design. The research design is a frame – like a skeleton – that holds together other research components including methods of data collection, sampling, and analysis. A research design framework can be experimental or non-experimental/descriptive. Most actual research projects combine elements of one or more designs.

Experimental designs are typically used when you are trying to test or establish *cause-effect relationships.*

Non-experimental/descriptive studies often answer "what is" or "what was" questions, and are *exploratory* in nature. Cross-sectional and longitudinal studies are two common varieties of this design. c

Cross-sectional studies provide a snapshot of the characteristics of a unit of the studied population in a given time period

Longitudinal studies observe and study changes in the unit of analysis over time.

Case studies focus on an object or phenomenon ("case"), whether it is one individual, one organization or its sub-unit, or one decision, etc., in order to provide an in-depth and rich account of that object or phenomenon in its context

Specify how you will sample your data sources

Sampling is a process of selecting a number of units from a defined population. Sampling can be classified into random/probability or non-random/nonprobability methods with further variations within each category

Select data collection methods

Data Collection Methods

Method	Use when	Advantages	Disadvantages
Document Review	Program documents or literature are available and can provide insight into the program or the evaluation	• Data already exist • Does not interrupt the program • Little or no burden on others • Can provide historical or comparison data • Introduces little bias	• Time consuming • Data limited to what exists and is available • Data may be incomplete • Requires clearly defining the data you're seeking
Observation	You want to learn how the program actually operates—its processes and activities	• Allows you to learn about the program as it is occurring • Can reveal unanticipated information of value • Flexible in the course of collecting data	• Time consuming • Having an observer can alter events • Difficult to observe multiple processes simultaneously • Can be difficult to interpret observed behaviors
Survey	You want information directly from a defined group of people to get a general idea of a situation, to generalize about a population, or to get a total count of a particular characteristic	• Many standardized instruments available • Can be anonymous • Allows a large sample • Standardized responses easy to analyze • Able to obtain a large amount of data quickly • Relatively low cost • Convenient for respondents	• Sample may not be representative • May have low return rate • Wording can bias responses • Closed-ended or brief responses may not provide the "whole story" • Not suited for all people—e.g., those with low reading level
Interview	You want to understand impressions and experiences in more detail and be able to expand or clarify responses	• Often better response rate than surveys • Allows flexibility in questions/probes • Allows more in-depth information to be gathered	• Time consuming • Requires skilled interviewer • Less anonymity for respondent • Qualitative data more difficult to analyze
Focus Group	You want to collect in-depth information from a group of people about their experiences and perceptions related to a specific issue.	• Collect multiple peoples' input in one session • Allows in-depth discussion • Group interaction can produce greater insight • Can be conducted in short time frame • Can be relatively inexpensive compared to interviews	• Requires skilled facilitator • Limited number of questions can be asked • Group setting may inhibit or influence opinions • Data can be difficult to analyze • Not appropriate for all topics or populations

Fig 5: Data Collection methods

Collect, Analyze, and Interpret Data

A typical report usually includes the following sections:

Executive summary with key points on what you did and what you found.

Introduction explaining what this research does and why.

Background providing relevant and necessary information for the reader
Findings and recommendations usually organized by questions and answering key questions
Appendices (methodology, references, etc.)

Draw practical recommendations for the client
The recommendations answer "*now what"* questions and specify how a client should act on the research findings

For each recommendation, clearly explain:

i. What this recommendation involves in some detail: What should your client/organization do based on your research findings?

ii. Why it is important: provide a rationale for your recommendation by referring to: (1) the original problem and its consequences that prompted the research project; and (2) challenges, opportunities, and strengths you have identified through this research. Use robust evidence and make your reasoning explicit. This information can motivate the client to act on your recommendations.

iii. How might the recommendation be implemented (and what should the client be wary of). Such implementation details can help your client to get started and make it easier to act accordingly. Otherwise, recommendations may stay only on paper.

Summary

Applied Research Process Guidelines in Brief

Criteria for good applied research design:

1. **Validity** (construct, statistical, internal, and external): is your study free of biases, distortions, and unsupported assumptions and conclusions?
2. **Reliability**: if someone else did this study would they arrive at similar findings?
3. **Effectiveness:** does your study meet its stated goals?
4. **Efficiency**: does this study use as little time, resources, and effort as possible to meet your goals and objectives?
5. **Feasibility**: is your study feasible for the time, expertise, and resources within which you would operate?
6. **Relevance:** is the information collected relevant for answering your key questions?
7. **Sufficiency**: is the information collected sufficient to answer your key questions?

Step 1.Clarify Your Research Focus Identify the problem/opportunity the research project will address

- Define the goal(s) and objectives of the research project
- Clarify logistics including by asking the following questions: How much time, resources, and access to information will you have for your project?
- What format will your deliverable follow: will you submit a report or a presentation or both? Will you build a website or another deliverable? Which format will best suit the project goal?
- Will you only report findings or will you also develop recommendations and/or an implementation plan?

Step 2.Scan Existing Information Conduct preliminary review of relevant information, including consulting relevant documents, academic literature, and key informants/experts. If needed, also visit relevant sites to observe the phenomenon in question.

Define key terms.

Articulate your conceptual framework/model specifying what might be happening with the phenomenon in question.

Revisit Step 1's research focus elements including the problem, goals, and objectives, if needed.

Articulate (or revisit) research questions typically corresponding to the goal and objectives respectively.

Define the scope and boundaries of the research (time, space, group, and other boundaries) articulating what is within and beyond your research.

Run items #1-6 by your client one more time to make sure you are on the same page.

Step 3.Plan Your Research Tasks and Methods

Select the research design framework (experimental or descriptive).

Specify the information needed to answer each research question.

Identify data sources for each piece of information needed.

Specify how you will sample your data sources.

(Specify variables and measurements if doing a quantitative study).

Select data collection methods.

Design data collection instruments/forms/questionnaires and procedures corresponding to each data collection method if you plan on using interviews, surveys, focus groups, or observations. Pilot your data collection instruments and revise them as needed.

Identify data analysis methods and tools for qualitative and/or quantitative data as relevant. Specify data collection, coding, and analysis procedures.

Plan how you will address validity and reliability issues.

Develop schedule (e.g. using a Gantt chart) for carrying out the tasks listed in this step.

(If doing formal proposal add a budget and information about the research team's qualifications.)

Step 4.Collect, analyze, and interpret the data Organize the information using a sample report outline in appendix #3 to free your brain for heavy lifting.

Collect data.

Analyze the data.

Draw practical recommendations for the client.

Step5. Revise, Polish, and Share Your Work Deliver final presentation.

Submit your written report.

Conclusion:

Once the general framework has been developed this can be used in the Domain framework like in the agricultural domain general or common section content for the final Applied Research report

References:

[1]: Mahabat Baimyrzaeva, Beginners' Guide for Applied Research, University of Central Asia, 2018